

Network Traffic & Address Resolution Audit

Evaluating Data Transit and Table Population

The Routine Audit Scenario

The organization has just installed a new central gateway appliance to connect the General Staff area and the Secure Records area. To ensure the devices are communicating correctly, the IT Audit team must perform a basic connectivity and discovery check. The team’s objective is simple: verify that standard data successfully travels between the two separate areas. More importantly, the team needs to observe how the network devices automatically discover one another and document how their internal address tables are populated with physical hardware addresses as the communication takes place.

Audit Guidelines

Simulation Setup

Build a virtual network representing the office environment. You will need two distinct network areas connected by one central Layer 3 gateway appliance. Ensure all device address tables are initially empty before beginning the tests.

IP Configuration Table

Zone	Device Role	IP Address	Default Gateway
General Staff Area	Audit Team Laptop	10.10.10.50	10.10.10.1
	Staff Workstation	10.10.10.60	10.10.10.1
Secure Records Area	Database Server	20.20.20.99	20.20.20.1
	File Server	20.20.20.100	20.20.20.1

Audit Tasks

Enable Simulation Mode and execute the following testing steps:

- Task 1:** *Local Table Population:* Send a standard data packet from the Audit Team Laptop to the Staff Workstation. Inspect the local switch and document how its physical address table dynamically populates with the addresses of both devices.
- Task 2:** *Gateway Discovery:* Send a data packet from the Audit Team Laptop to the Database Server in the other area. Observe the initial broadcast request generated by the laptop to discover the gateway’s physical address.
- Task 3:** *Gateway Transit:* Let the packet from Task 2 reach the central gateway. Pause the simulation. Open the packet details and inspect how the gateway changes the Data Link Layer (Layer 2) addresses before sending it into the Secure Records Area.

Final Report Submission**Deliverables**

Compile your findings into a single PDF audit report containing:

1. The simulation save file demonstrating the configured network areas.
2. Screenshots of the local switch's populated address table after Task 1 is completed.
3. A brief, plain-text explanation of the exact changes the central gateway applies to the physical hardware addresses as data moves across the boundary in Task 3.